ICE BOX — SECTION

SCALE 1:5

EVAPORATOR AND BOX DATA	
evaporating temperature	-5 C
evaporating pressure	2.92 kPa abs
cooling delivered per night	2.19 MJ
mean power over 12 h	**51 W**
surface required	**508 cm2**
box internal	500 x 350 x 300
insulation U	0.30 W/m2K
heat leak at 30 C ambient	8 W
leak as share of the night	**15 %**

NOTES

1. The evaporator runs at 2.9 kPa absolute. It sees full external atmospheric pressure whenever the machine is cold - design it as a vacuum vessel, not a pressure vessel.
2. Plate-coil in SS 316L, all welded, no fittings inside the box.
3. Evaporator sits ABOVE the trays so liquid methanol drains back to the receiver by gravity when the machine warms.
4. Secondary containment: the box is lined with a welded SS tray connected to the bund on SI-300. A leak must not reach the ice.
5. The ice is NOT potable. Methanol is on the other side of one wall. Use the ice for cooling, not for drinks.
6. 80 mm PU gives a 15 % overnight leak, already inside the 5 kg figure.

INSPECTION		
CHECK	**REQUIREMENT**	**METHOD**
Evaporator external collapse	full vacuum, no set	1 bar external, 1 h
Helium leak	< 1e-8 Pa.m3/s	mass spectrometer
Containment tray	welded, drains to bund	water fill test
Box heat leak	<= 10 W at 30 C	calibrated pull-down

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **EVAPORATOR AND ICE BOX**
5 kg ICE PER NIGHT / 51 W MEAN / 80 mm PU

SCALE: **1:5** — THIRD ANGLE PROJECTION
UNITS: **mm**

DRAWING No. **SI-104**
REV **A**
SHEET **4 / 6**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-15